

Workplace Mental Health & Productivity Analytics

A Data Analytics Portfolio Project

MySQL • Python (Pandas, Matplotlib, Seaborn) • Streamlit • EDA

Author: Data Analytics Beginner | **Journey Started:** ~10 months ago | **Dataset:** mental_health.mental_health_workplace | **Records:** 10,000+

01 | ABOUT THIS PROJECT

I started learning data analytics about 10 months ago — completely from scratch. This project is my most complete end-to-end work so far. I took a real workplace mental health dataset and went through the full analytics workflow: cleaning and exploring the data with Python, writing SQL queries to answer business questions, building charts to communicate the findings, and finally creating an interactive Streamlit dashboard so anyone can explore the data themselves.

The dataset covers 10,000+ employee survey responses across 15+ countries and 5 years (2020-2024), looking at how work environments, overtime, and company mental health policies affect burnout, productivity, and wellbeing.

02 | TOOLS & TECHNOLOGIES USED

MySQL

Python

Pandas

Matplotlib

Seaborn

Streamlit

MySQL

Wrote 13 SQL queries using GROUP BY, CASE WHEN, AVG/SUM/COUNT, subqueries, and conditional logic to answer HR business questions.

Pandas

Used for data loading, cleaning, null checks, type conversion, and exploratory data analysis (EDA) — .describe(), .value_counts(), .corr().

Matplotlib & Seaborn

Created 13 visualisations including bar charts, violin plots, pie charts, and bubble charts to communicate findings visually.

Streamlit

Built an interactive dashboard with filters by country, industry and work model — letting users explore the data themselves in real time.

GitHub

All code, queries, and visualisations published as a public portfolio project for recruiters and the data community.

03 | EXPLORATORY DATA ANALYSIS (EDA) & DASHBOARD

What I did in EDA

- Loaded the dataset and checked shape, dtypes, and nulls
- Used `.describe()` to understand score distributions
- Plotted histograms of burnout risk and salary
- Created Seaborn heatmaps to find correlations
- Used `.value_counts()` on work model and gender columns
- Identified overtime vs non-overtime subgroups for comparison

What the Streamlit Dashboard Does

- Sidebar filters: country, industry, work model
- KPI cards showing avg burnout, salary, and EAP rate
- Dynamic charts that update when filters change
- Overtime impact section with side-by-side comparison
- Industry risk profile table sorted by burnout score
- Deployed as a shareable web app for GitHub portfolio

04 | SQL ANALYSIS — 13 BUSINESS QUESTIONS ANSWERED

Q1	Job role breakdown for 2024 — who makes up the workforce?
Q2	Nurses: average burnout risk score vs weekly work hours
Q3	Which industry pays the highest average salary?
Q4	Survey participation trend year-by-year (2020–2024)
Q5	Top 10 most represented countries in the study
Q6	EAP utilisation rate by country (used vs available)
Q7	Identifying high-overtime job roles by average weekly hours
Q8	Base salary statistics split by gender
Q9	Work model distribution: Remote vs Hybrid vs On-site
Q10	Burnout risk and work-life balance: overtime vs no-overtime
Q11	Industry mental health risk profiles: burnout + productivity + absenteeism
Q12	Absenteeism days by mental health condition
Q13	Do mental health policies improve job satisfaction and productivity?

05 | KEY FINDINGS & INSIGHTS

■ 121% Higher Burnout with Overtime	Employees working overtime score 4.83 on burnout vs only 2.18 for non-overtime workers. Work-life balance also drops from 6.64 to 6.35. A clear signal for HR teams.
■ Technology = Highest Salary (\$109K avg)	Tech leads all industries on average annual salary, followed by Legal (\$107K) and Finance & Banking (\$103K). Hospitality earns the least at \$54K.
■ Finance & Banking = Highest Burnout (4.94)	Despite high salaries, Finance & Banking tops the burnout risk table. Consulting (4.93) and Technology (4.89) follow closely — high pay, high pressure.

■ Sweden Leads EAP Utilisation at 11.6%	Sweden has the highest rate of employees actually using their Employee Assistance Programme, suggesting stronger mental health culture vs countries like Japan (9.3%).
■ Hybrid is the Most Common Work Model (35.7%)	Hybrid slightly edges out On-site (34.4%) with Remote at 29.9%. The post-pandemic shift to flexible working is clearly reflected in this 2020-2024 data.
■ Mental Health Policies Improve Satisfaction	Companies with a mental health policy score 5.90 on job satisfaction vs 5.85 without. Small margin — but across 4,259 employees, it adds up to a meaningful HR investment.

06 | PROJECT AT A GLANCE

10K+	13	13	15+	5 yrs	4
Employee Records Analysed	SQL Queries Written	Charts & Visualisations	Countries Covered	Data Range 2020-2024	Tools Used End-to-End

07 | A NOTE ON MY LEARNING JOURNEY

I started learning data analytics completely from scratch just 10 months ago. Before this, I had no background in SQL, Python, or data visualisation. This project represents my biggest milestone so far — it is the first time I have combined SQL, Python EDA, data visualisation, and a live Streamlit dashboard all in a single workflow. Every query, chart, and line of code in this project was written and debugged by me as part of my learning process. I built this to prove to myself (and to potential employers) that I can take a raw dataset and turn it into real, actionable insights — which is exactly what data analysts do every day.

08 | HOW TO RUN THIS PROJECT

1. Clone repo	git clone https://github.com/YOUR_USERNAME/workplace-mental-health-analytics
2. Install deps	pip install pandas matplotlib seaborn streamlit
3. Run EDA	Open eda_analysis.ipynb in Jupyter Notebook or VS Code
4. Run SQL	Import the dataset into MySQL and run queries from /sql/queries.sql
5. Launch app	streamlit run app.py (opens the interactive dashboard in your browser)